

Norway

SLEIPNER

Volve F

F-12

F-12

Design: F-12

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Actual RTE as installed)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Actual RTE as installed)
Well:	F-12	North Reference:	Grid
Wellbore:	F-12	Survey Calculation Method:	Minimum Curvature
Design:	F-12	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site		Volve F, 15/9				
Site Position:		Northing:	6,478,563.53 m	Latitude:	58° 26' 29.8068 N	
From:	Map	Easting:	435,050.02 m	Longitude:	1° 53' 14.9285 E	
Position Uncertainty:		0.00 m	Slot Radius:	13.200 in	Grid Convergence:	-0.95 °

Well	F-12					
Well Position	+N/-S	2.70 m	Northing:	6,478,566.22 m	Latitude:	58° 26' 29.8941 N
	+E/-W	0.19 m	Easting:	435,050.21 m	Longitude:	1° 53' 14.9374 E
Position Uncertainty		0.00 m	Wellhead Depth:	91.00 m	Water Depth:	91.00 m

Wellbore	F-12				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	MNETICREFERENCE	01.03.2007	-2.34	71.63	50,321

Design	F-12				
Audit Notes:					
Version:	1.0	Phase:	ACTUAL	Tie On Depth:	145.90
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)	
	145.90	2.70	0.19	104.32	

Survey Program	Date	30.09.2017			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
174.00	2,464.55	14" Casing Gyro Survey (F-12)	NOTuse_Wellbore Surveyor,	Gyro Tool from GD	
2,465.55	2,536.07	17.5" Section MWD Survey (F-12)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)	
2,536.07	3,081.25	12.25" Section MWD Survey (F-12)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)	
3,136.60	3,495.52	8.5" Section MWD Survey (F-12)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)	

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
145.90	0.00	0.00	145.90	2.70	0.19	0.00	0.000	0.000	0.000	
174.00	0.12	166.21	174.00	2.67	0.20	0.01	0.132	0.132	0.000	
204.00	0.08	159.79	204.00	2.62	0.21	0.04	0.045	-0.043	-6.416	
234.00	0.15	113.27	234.00	2.58	0.25	0.09	0.109	0.067	-46.518	
264.00	0.26	94.15	264.00	2.56	0.36	0.20	0.125	0.108	-19.124	
294.00	0.34	83.77	294.00	2.57	0.51	0.34	0.101	0.086	-10.379	
324.00	0.36	81.51	324.00	2.59	0.70	0.52	0.027	0.023	-2.257	
354.00	0.46	79.36	354.00	2.63	0.91	0.71	0.101	0.100	-2.154	

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Design:	F-12	Database:	Production EDM P246N

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Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
384.00	0.65	66.92	384.00	2.72	1.19	0.96	0.224	0.189	-12.439
414.00	1.09	38.86	413.99	3.01	1.52	1.22	0.597	0.435	-28.058
444.00	2.91	27.73	443.97	3.90	2.06	1.51	1.856	1.824	-11.131
474.00	4.85	26.01	473.90	5.72	2.97	1.94	1.937	1.934	-1.720
504.00	6.49	27.94	503.76	8.35	4.32	2.60	1.657	1.646	1.930
534.00	7.69	30.42	533.53	11.58	6.13	3.56	1.234	1.196	2.473
564.00	6.95	32.03	563.28	14.85	8.11	4.66	0.771	-0.743	1.617
594.00	5.54	30.06	593.10	17.64	9.79	5.61	1.419	-1.403	-1.976
624.00	4.90	17.77	622.98	20.12	10.91	6.08	1.286	-0.642	-12.284
654.00	5.08	0.38	652.87	22.66	11.31	5.84	1.518	0.180	-17.395
684.00	5.95	334.67	682.73	25.40	10.65	4.52	2.596	0.874	-25.711
714.00	6.39	320.18	712.56	28.09	8.92	2.18	1.611	0.433	-14.485
744.00	6.80	303.74	742.36	30.36	6.37	-0.85	1.924	0.408	-16.445
774.00	8.44	288.20	772.10	32.03	2.80	-4.72	2.621	1.644	-15.531
804.00	11.14	278.45	801.66	33.14	-2.15	-9.80	3.159	2.700	-9.755
834.00	13.14	272.24	830.99	33.70	-8.43	-16.02	2.388	2.003	-6.205
864.00	14.81	266.26	860.11	33.59	-15.66	-23.00	2.204	1.667	-5.984
893.00	14.99	264.04	888.13	32.95	-23.09	-30.04	0.620	0.186	-2.300
903.00	14.51	263.13	897.80	32.67	-25.62	-32.42	1.599	-1.442	-2.717
913.00	14.08	262.05	907.49	32.35	-28.07	-34.72	1.513	-1.283	-3.250
923.00	13.81	261.01	917.20	32.00	-30.45	-36.94	1.096	-0.799	-3.112
933.00	13.78	260.55	926.91	31.62	-32.81	-39.13	0.347	-0.110	-1.382
943.00	13.86	260.53	936.62	31.22	-35.16	-41.31	0.251	0.250	-0.060
953.00	13.96	260.58	946.33	30.83	-37.54	-43.51	0.301	0.299	0.138
963.00	14.13	261.38	956.03	30.45	-39.93	-45.74	0.771	0.505	2.403
973.00	14.43	263.23	965.72	30.12	-42.38	-48.03	1.649	0.913	5.563
983.00	14.70	264.88	975.40	29.86	-44.88	-50.39	1.475	0.794	4.942
993.00	14.91	265.80	985.06	29.65	-47.43	-52.80	0.956	0.640	2.778
1,003.00	15.04	266.46	994.72	29.48	-50.00	-55.26	0.625	0.370	1.952
1,013.00	15.13	267.49	1,004.38	29.34	-52.60	-57.74	0.855	0.273	3.113
1,023.00	15.14	268.09	1,014.03	29.24	-55.21	-60.24	0.468	0.048	1.781
1,033.00	15.07	268.61	1,023.69	29.16	-57.82	-62.75	0.470	-0.230	1.573
1,043.00	14.54	269.28	1,033.36	29.12	-60.37	-65.21	1.664	-1.583	2.005
1,053.00	14.23	269.64	1,043.04	29.09	-62.85	-67.61	0.972	-0.933	1.089
1,063.00	14.15	269.65	1,052.74	29.08	-65.30	-69.98	0.232	-0.232	0.011
1,073.00	14.22	269.22	1,062.43	29.05	-67.75	-72.35	0.385	0.220	-1.288
1,083.00	14.38	267.93	1,072.12	28.99	-70.22	-74.73	1.061	0.466	-3.858
1,093.00	14.59	265.82	1,081.81	28.85	-72.72	-77.11	1.712	0.650	-6.332
1,103.00	14.67	263.48	1,091.48	28.62	-75.24	-79.49	1.787	0.236	-7.010
1,113.00	14.82	260.96	1,101.15	28.27	-77.76	-81.85	1.977	0.434	-7.579
1,123.00	15.14	258.59	1,110.81	27.81	-80.30	-84.20	2.077	0.976	-7.092
1,133.00	15.71	256.39	1,120.45	27.24	-82.90	-86.57	2.443	1.697	-6.610
1,143.00	16.43	254.47	1,130.06	26.54	-85.57	-89.00	2.685	2.162	-5.750

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Well:	F-12	North Reference:	Grid
Wellbore:	F-12	Survey Calculation Method:	Minimum Curvature
Design:	F-12	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
1,153.00	17.58	252.27	1,139.62	25.70	-88.38	-91.50	3.945	3.442	-6.600
1,163.00	18.82	250.54	1,149.12	24.71	-91.33	-94.12	4.059	3.723	-5.185
1,173.00	19.84	248.95	1,158.56	23.56	-94.44	-96.85	3.456	3.073	-4.783
1,183.00	20.52	248.22	1,167.95	22.30	-97.65	-99.65	2.169	2.032	-2.195
1,193.00	20.78	248.08	1,177.30	20.99	-100.92	-102.50	0.797	0.784	-0.416
1,203.00	20.87	248.00	1,186.65	19.66	-104.22	-105.36	0.295	0.281	-0.248
1,213.00	20.57	248.50	1,196.00	18.34	-107.51	-108.22	1.047	-0.900	1.515
1,223.00	19.87	249.42	1,205.39	17.10	-110.73	-111.04	2.308	-2.103	2.753
1,233.00	19.22	250.72	1,214.81	15.96	-113.88	-113.81	2.349	-1.955	3.892
1,243.00	18.88	251.85	1,224.26	14.91	-116.97	-116.54	1.507	-1.019	3.400
1,253.00	18.83	252.46	1,233.73	13.92	-120.05	-119.28	0.608	-0.151	1.823
1,263.00	19.07	252.60	1,243.18	12.95	-123.15	-122.04	0.736	0.724	0.414
1,273.00	19.35	252.60	1,252.63	11.96	-126.29	-124.84	0.842	0.841	0.022
1,283.00	19.47	252.47	1,262.06	10.97	-129.46	-127.66	0.385	0.359	-0.411
1,293.00	19.09	252.04	1,271.50	9.96	-132.60	-130.46	1.223	-1.150	-1.266
1,303.00	17.98	251.23	1,280.98	8.96	-135.62	-133.14	3.404	-3.315	-2.432
1,313.00	16.56	250.46	1,290.53	7.99	-138.42	-135.61	4.329	-4.273	-2.319
1,323.00	15.24	250.09	1,300.14	7.06	-141.00	-137.88	3.970	-3.958	-1.119
1,333.00	14.36	250.04	1,309.81	6.19	-143.40	-140.00	2.658	-2.657	-0.130
1,343.00	14.09	249.89	1,319.51	5.35	-145.71	-142.02	0.791	-0.782	-0.477
1,353.00	14.20	248.89	1,329.20	4.49	-148.00	-144.03	0.796	0.318	-2.988
1,363.00	14.61	247.86	1,338.89	3.57	-150.31	-146.04	1.450	1.229	-3.088
1,373.00	15.35	247.18	1,348.55	2.58	-152.70	-148.11	2.271	2.210	-2.028
1,383.00	16.31	246.75	1,358.17	1.52	-155.21	-150.28	2.925	2.904	-1.299
1,393.00	17.11	246.17	1,367.75	0.37	-157.85	-152.55	2.429	2.377	-1.743
1,403.00	17.79	245.40	1,377.29	-0.86	-160.58	-154.89	2.168	2.056	-2.296
1,413.00	18.50	244.47	1,386.79	-2.18	-163.40	-157.30	2.294	2.119	-2.818
1,423.00	19.40	243.27	1,396.25	-3.61	-166.32	-159.77	2.945	2.704	-3.589
1,433.00	20.36	242.01	1,405.65	-5.18	-169.34	-162.31	3.154	2.880	-3.781
1,443.00	21.22	240.96	1,415.00	-6.87	-172.45	-164.91	2.809	2.578	-3.140
1,453.00	22.14	239.94	1,424.29	-8.69	-175.67	-167.58	2.979	2.754	-3.076
1,463.00	22.83	239.04	1,433.53	-10.64	-178.96	-170.29	2.307	2.066	-2.682
1,473.00	22.85	238.62	1,442.75	-12.64	-182.28	-173.01	0.500	0.068	-1.276
1,483.00	22.44	238.58	1,451.98	-14.65	-185.57	-175.70	1.242	-1.241	-0.112
1,493.00	21.71	239.22	1,461.24	-16.59	-188.79	-178.33	2.303	-2.185	1.934
1,503.00	21.01	240.26	1,470.56	-18.43	-191.93	-180.93	2.370	-2.080	3.121
1,513.00	20.67	241.23	1,479.90	-20.17	-195.04	-183.50	1.452	-1.022	2.899
1,523.00	20.66	242.19	1,489.26	-21.84	-198.14	-186.10	1.013	-0.051	2.866
1,533.00	20.83	243.57	1,498.61	-23.45	-201.30	-188.76	1.554	0.519	4.137
1,543.00	20.97	245.58	1,507.95	-24.98	-204.52	-191.50	2.202	0.432	6.053
1,553.00	21.13	247.64	1,517.29	-26.41	-207.81	-194.34	2.269	0.478	6.175
1,563.00	21.38	249.38	1,526.61	-27.74	-211.19	-197.28	2.028	0.732	5.218
1,573.00	21.64	251.07	1,535.91	-28.98	-214.64	-200.32	2.027	0.805	5.072
1,583.00	21.87	252.47	1,545.20	-30.14	-218.16	-203.44	1.689	0.663	4.192

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1,593.00	22.08	253.27	1,554.47	-31.24	-221.73	-206.63	1.110	0.652	2.402
1,603.00	22.29	253.50	1,563.73	-32.32	-225.35	-209.87	0.675	0.621	0.703
1,613.00	22.47	253.52	1,572.98	-33.40	-229.00	-213.14	0.555	0.554	0.056
1,623.00	22.66	253.41	1,582.21	-34.49	-232.68	-216.44	0.576	0.561	-0.340
1,633.00	22.93	253.25	1,591.43	-35.60	-236.39	-219.76	0.820	0.799	-0.472
1,643.00	23.20	253.11	1,600.63	-36.74	-240.14	-223.11	0.832	0.816	-0.424
1,653.00	23.40	252.96	1,609.82	-37.89	-243.93	-226.49	0.616	0.590	-0.448
1,663.00	23.53	252.89	1,618.99	-39.06	-247.73	-229.89	0.415	0.405	-0.229
1,673.00	23.59	252.88	1,628.16	-40.24	-251.55	-233.30	0.179	0.178	-0.023
1,683.00	23.66	252.91	1,637.32	-41.41	-255.38	-236.72	0.199	0.195	0.102
1,693.00	23.70	252.85	1,646.48	-42.60	-259.22	-240.15	0.141	0.118	-0.192
1,703.00	23.73	252.87	1,655.63	-43.78	-263.06	-243.58	0.102	0.098	0.068
1,713.00	23.78	252.93	1,664.78	-44.97	-266.91	-247.01	0.164	0.147	0.182
1,723.00	23.76	252.96	1,673.94	-46.15	-270.77	-250.46	0.057	-0.044	0.089
1,733.00	23.71	252.97	1,683.09	-47.33	-274.62	-253.89	0.145	-0.145	0.020
1,743.00	23.70	253.01	1,692.25	-48.50	-278.46	-257.33	0.070	-0.049	0.126
1,753.00	23.68	253.05	1,701.40	-49.68	-282.30	-260.76	0.065	-0.041	0.125
1,763.00	23.63	252.90	1,710.56	-50.85	-286.14	-264.19	0.252	-0.169	-0.466
1,773.00	23.56	252.97	1,719.73	-52.03	-289.97	-267.60	0.221	-0.203	0.216
1,783.00	23.50	252.85	1,728.90	-53.20	-293.78	-271.01	0.235	-0.188	-0.350
1,793.00	23.40	252.68	1,738.07	-54.38	-297.58	-274.40	0.358	-0.296	-0.505
1,803.00	23.37	252.68	1,747.25	-55.56	-301.37	-277.78	0.100	-0.100	-0.004
1,813.00	23.36	252.63	1,756.43	-56.74	-305.16	-281.16	0.063	-0.009	-0.156
1,823.00	23.39	252.65	1,765.61	-57.93	-308.94	-284.53	0.094	0.091	0.058
1,833.00	23.45	252.52	1,774.78	-59.12	-312.74	-287.91	0.241	0.186	-0.385
1,843.00	23.48	252.56	1,783.96	-60.31	-316.54	-291.30	0.089	0.075	0.120
1,853.00	23.41	252.72	1,793.13	-61.50	-320.33	-294.69	0.294	-0.219	0.494
1,863.00	23.39	252.72	1,802.31	-62.68	-324.12	-298.07	0.062	-0.062	-0.016
1,873.00	23.47	252.78	1,811.48	-63.86	-327.92	-301.46	0.258	0.248	0.185
1,883.00	23.51	252.91	1,820.66	-65.03	-331.73	-304.85	0.195	0.122	0.379
1,893.00	23.49	253.11	1,829.83	-66.20	-335.54	-308.26	0.247	-0.056	0.604
1,903.00	23.50	253.09	1,839.00	-67.36	-339.36	-311.67	0.033	0.024	-0.055
1,913.00	23.55	253.15	1,848.17	-68.51	-343.18	-315.08	0.162	0.144	0.189
1,923.00	23.58	253.33	1,857.33	-69.67	-347.00	-318.51	0.232	0.094	0.531
1,933.00	23.56	253.55	1,866.50	-70.81	-350.84	-321.94	0.271	-0.044	0.668
1,943.00	23.58	253.60	1,875.66	-71.94	-354.67	-325.38	0.077	0.048	0.149
1,953.00	23.57	253.53	1,884.83	-73.07	-358.51	-328.81	0.086	-0.017	-0.210
1,963.00	23.19	253.55	1,894.01	-74.19	-362.32	-332.22	1.155	-1.154	0.050
1,973.00	22.48	253.50	1,903.22	-75.29	-366.04	-335.56	2.121	-2.120	-0.161
1,983.00	21.77	253.27	1,912.49	-76.37	-369.65	-338.79	2.143	-2.128	-0.667
1,993.00	21.15	252.84	1,921.79	-77.44	-373.15	-341.92	1.931	-1.872	-1.299
2,003.00	20.52	252.07	1,931.14	-78.51	-376.54	-344.94	2.049	-1.875	-2.322
2,013.00	19.83	251.05	1,940.53	-79.60	-379.81	-347.84	2.324	-2.074	-3.041

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Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Actual RTE as installed)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Actual RTE as installed)
Well:	F-12	North Reference:	Grid
Wellbore:	F-12	Survey Calculation Method:	Minimum Curvature
Design:	F-12	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
2,023.00	19.24	249.89	1,949.95	-80.72	-382.96	-350.62	2.120	-1.771	-3.487	
2,033.00	18.67	248.97	1,959.41	-81.86	-386.00	-353.28	1.940	-1.722	-2.753	
2,043.00	18.09	248.27	1,968.90	-83.01	-388.94	-355.84	1.867	-1.744	-2.116	
2,053.00	17.42	247.76	1,978.42	-84.15	-391.77	-358.30	2.062	-2.009	-1.519	
2,063.00	16.64	247.57	1,987.98	-85.26	-394.48	-360.65	2.329	-2.323	-0.585	
2,073.00	15.74	247.32	1,997.59	-86.33	-397.05	-362.88	2.730	-2.722	-0.735	
2,083.00	14.82	246.86	2,007.23	-87.36	-399.48	-364.98	2.786	-2.761	-1.398	
2,093.00	14.11	246.33	2,016.92	-88.35	-401.77	-366.95	2.142	-2.105	-1.586	
2,103.00	13.55	245.78	2,026.63	-89.32	-403.96	-368.83	1.739	-1.694	-1.632	
2,113.00	12.56	245.14	2,036.37	-90.26	-406.01	-370.59	3.005	-2.973	-1.935	
2,123.00	11.76	244.30	2,046.14	-91.15	-407.92	-372.21	2.462	-2.404	-2.526	
2,133.00	11.30	243.19	2,055.94	-92.04	-409.71	-373.73	1.514	-1.360	-3.319	
2,143.00	10.90	242.00	2,065.75	-92.92	-411.42	-375.17	1.382	-1.198	-3.575	
2,153.00	10.37	240.37	2,075.58	-93.81	-413.04	-376.52	1.841	-1.604	-4.889	
2,163.00	9.96	237.45	2,085.43	-94.72	-414.55	-377.75	1.978	-1.233	-8.767	
2,173.00	9.68	233.80	2,095.28	-95.69	-415.95	-378.88	2.040	-0.833	-10.924	
2,183.00	9.51	229.85	2,105.14	-96.72	-417.26	-379.89	2.044	-0.512	-11.877	
2,193.00	9.30	225.13	2,115.00	-97.82	-418.47	-380.79	2.398	-0.643	-14.143	
2,203.00	9.20	220.24	2,124.87	-99.00	-419.56	-381.55	2.375	-0.291	-14.673	
2,213.00	9.10	215.05	2,134.75	-100.25	-420.53	-382.18	2.491	-0.301	-15.555	
2,223.00	9.02	209.83	2,144.62	-101.58	-421.37	-382.67	2.481	-0.241	-15.689	
2,233.00	8.95	204.57	2,154.50	-102.97	-422.08	-383.02	2.471	-0.197	-15.780	
2,243.00	8.91	199.61	2,164.38	-104.41	-422.67	-383.23	2.309	-0.111	-14.857	
2,253.00	8.98	195.16	2,174.26	-105.89	-423.13	-383.31	2.088	0.193	-13.370	
2,263.00	9.11	189.99	2,184.13	-107.42	-423.47	-383.26	2.472	0.405	-15.512	
2,273.00	9.31	184.67	2,194.01	-109.01	-423.68	-383.07	2.617	0.590	-15.934	
2,283.00	9.55	179.31	2,203.87	-110.65	-423.73	-382.72	2.739	0.733	-16.108	
2,293.00	9.88	174.47	2,213.73	-112.33	-423.64	-382.21	2.638	0.986	-14.503	
2,303.00	10.25	171.03	2,223.57	-114.06	-423.42	-381.57	2.116	1.104	-10.332	
2,313.00	10.48	167.42	2,233.41	-115.83	-423.08	-380.80	2.062	0.681	-10.818	
2,323.00	10.69	163.89	2,243.24	-117.61	-422.63	-379.92	2.052	0.644	-10.607	
2,333.00	10.99	161.52	2,253.06	-119.40	-422.07	-378.94	1.612	0.903	-7.100	
2,343.00	11.41	158.88	2,262.87	-121.23	-421.41	-377.85	1.974	1.237	-7.919	
2,353.00	11.77	156.24	2,272.67	-123.09	-420.64	-376.64	1.934	1.103	-7.911	
2,363.00	12.25	153.98	2,282.45	-124.97	-419.76	-375.33	2.011	1.431	-6.791	
2,373.00	12.79	151.93	2,292.21	-126.90	-418.78	-373.89	2.098	1.620	-6.154	
2,383.00	13.35	149.87	2,301.95	-128.88	-417.68	-372.34	2.184	1.681	-6.166	
2,393.00	14.04	147.62	2,311.67	-130.90	-416.45	-370.65	2.614	2.065	-6.769	
2,403.00	14.66	145.72	2,321.35	-132.97	-415.08	-368.82	2.328	1.855	-5.677	
2,413.00	15.34	143.81	2,331.01	-135.09	-413.59	-366.85	2.524	2.041	-5.736	
2,423.00	15.89	141.95	2,340.65	-137.23	-411.97	-364.74	2.237	1.655	-5.591	
2,433.00	16.32	140.28	2,350.25	-139.39	-410.22	-362.52	1.886	1.277	-5.005	
2,443.00	16.72	138.81	2,359.84	-141.55	-408.38	-360.20	1.741	1.209	-4.405	

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Actual RTE as installed)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Actual RTE as installed)
Well:	F-12	North Reference:	Grid
Wellbore:	F-12	Survey Calculation Method:	Minimum Curvature
Design:	F-12	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
2,453.00	17.17	136.78	2,369.41	-143.71	-406.42	-357.77	2.223	1.343	-6.078
2,463.00	17.67	134.92	2,378.95	-145.86	-404.34	-355.21	2.261	1.521	-5.592
2,464.55	17.75	134.64	2,380.42	-146.19	-404.00	-354.81	2.271	1.546	-5.464
2,493.74	18.80	129.52	2,408.14	-152.31	-397.21	-346.71	1.970	1.077	-5.261
2,536.07	20.39	125.56	2,448.02	-160.94	-385.94	-333.66	1.467	1.125	-2.807
2,576.99	21.70	125.92	2,486.21	-169.52	-374.02	-319.98	0.965	0.960	0.264
2,617.92	26.34	124.65	2,523.59	-179.13	-360.41	-304.42	3.422	3.401	-0.931
2,658.30	30.56	121.61	2,559.08	-189.61	-344.29	-286.21	3.314	3.135	-2.259
2,699.07	33.11	117.41	2,593.72	-200.17	-325.57	-265.47	2.485	1.876	-3.091
2,738.83	35.52	115.77	2,626.56	-210.20	-305.53	-243.56	1.947	1.818	-1.237
2,779.26	41.25	113.83	2,658.24	-220.70	-282.74	-218.89	4.344	4.252	-1.440
2,820.02	45.33	112.36	2,687.90	-231.64	-257.03	-191.27	3.093	3.003	-1.082
2,860.02	49.20	108.88	2,715.05	-241.96	-229.54	-162.08	3.478	2.902	-2.610
2,900.45	51.78	104.36	2,740.77	-250.85	-199.66	-130.93	3.218	1.914	-3.354
2,939.72	53.48	102.53	2,764.61	-258.10	-169.31	-99.73	1.709	1.299	-1.398
2,980.29	54.35	103.04	2,788.51	-265.36	-137.34	-66.95	0.712	0.643	0.377
3,008.61	54.95	103.40	2,804.89	-270.64	-114.85	-43.86	0.708	0.636	0.381
3,060.63	54.55	99.95	2,834.92	-279.24	-73.26	-1.43	1.641	-0.231	-1.990
3,081.25	54.06	97.21	2,846.95	-281.74	-56.70	15.23	3.315	-0.713	-3.986
3,136.60	53.84	95.71	2,879.53	-286.77	-12.24	59.56	0.668	-0.119	-0.813
3,172.51	53.90	98.14	2,900.70	-290.27	16.55	88.32	1.640	0.050	2.030
3,216.39	53.86	101.30	2,926.57	-296.25	51.48	123.64	1.745	-0.027	2.160
3,257.05	53.34	103.39	2,950.70	-303.25	83.45	156.34	1.299	-0.384	1.542
3,297.38	53.81	105.77	2,974.65	-311.42	114.85	188.79	1.467	0.350	1.770
3,337.63	53.10	107.03	2,998.62	-320.55	145.87	221.10	0.922	-0.529	0.939
3,375.41	53.01	106.59	3,021.33	-329.28	174.77	251.27	0.288	-0.071	-0.349
3,416.55	53.14	107.02	3,046.04	-338.79	206.26	284.13	0.268	0.095	0.314
3,456.15	52.97	106.74	3,069.84	-347.98	236.54	315.74	0.213	-0.129	-0.212
3,495.52	52.87	105.59	3,093.58	-356.72	266.71	347.14	0.703	-0.076	-0.876
3,520.00	52.87	105.59	3,108.36	-361.97	285.51	366.65	0.000	0.000	0.000

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Actual RTE as installed)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Actual RTE as installed)
Well:	F-12	North Reference:	Grid
Wellbore:	F-12	Survey Calculation Method:	Minimum Curvature
Design:	F-12	Database:	Production EDM P246N

Design Targets									
Target Name - hit/miss target - Shape	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude	Longitude
Start Tangent - actual wellpath misses target center by 24.50m at 3032.23m MD (2818.49 TVD, -274.86 N, -96.00 E) - Point	0.00	0.00	2,821.28	-250.70	-93.03	6,478,312.91	434,957.03	58° 26' 21.6557 N	1° 53' 9.4514 E
Volve P60 50deg Topp - actual wellpath misses target center by 17.39m at 3123.58m MD (2871.85 TVD, -285.70 N, -22.70 E) - Point	0.00	0.00	2,874.90	-268.58	-22.99	6,478,295.04	435,027.04	58° 26' 21.1155 N	1° 53' 13.7856 E
Volve P61 50deg Bunn - actual wellpath misses target center by 6.95m at 3293.78m MD (2972.52 TVD, -310.63 N, 112.05 E) - Point	0.00	0.00	2,976.90	-305.63	110.02	6,478,258.00	435,160.00	58° 26' 19.9893 N	1° 53' 22.0199 E

Casing Points						
Measured Depth (m)	Vertical Depth (m)	Name			Casing Diameter (in)	Hole Diameter (in)
251.00	251.00	30"			30.000	36.000
1,357.00	1,333.08	20"			20.000	26.000
2,506.00	2,419.73	14"			14.000	17.500
3,113.00	2,865.62	9 5/8"			9.625	12.250
3,519.00	3,107.75	7"			7.000	8.500

Design Annotations					
Measured Depth (m)	Vertical Depth (m)	Local Coordinates		Comment	
		+N/-S (m)	+E/-W (m)		
3,520.00	3,108.36	-361.97	285.51	Projection to TD	

Checked By: _____ Approved By: _____ Date: _____